

Project Rubric: Applications of Signals and Systems

Criteria	Weight	Excellent (4 pts)	Proficient (3 pts)	Developing (2 pts)	Beginning (1 pt)	Score
1. Problem Definition & Theoretical Modeling	15%	Insightful Application: - Identifies a clear, relevant, and non-trivial real-world problem (e.g., audio artifact removal, image edge-detection). - Correctly and insightfully maps the problem to S&S concepts (e.g., filtering, Fourier-transforms) and justifies	Clear Application: - Identifies a clear and relevant problem. - Correctly applies core S&S concepts with clear justification.	Basic Application: - Problem is overly simplistic, poorly defined, or a weak fit for S&S. - Applies some concepts correctly, but with errors or weak justification.	Lacks Application: - Fails to identify a valid S&S problem. - Theoretical model is missing or fundamentally flawed.	/ 15

		all design choices.				
2. Implementation & Code Quality	40%	Exemplary Code: - Code is efficient, well-structured, and clearly commented. Correctly implements the theoretical model. - Successfully processes real-world or realistic sample data (e.g., a real .wav file, a .jpg image).	Functional Code: - Code works and implements the core goals of the project. - Processes sample data correctly, though may have minor bugs or inefficiencies. Code is understandable.	Partially Functional Code: - Code runs but has significant bugs, fails to implement key requirements, or only works on "perfect" (trivial) test data. - Logic is hard to follow.	Non-Functional Code: - Code is incomplete, does not run, or is fundamentally incorrect. - Minimal effort is evident.	/ 40
3. Analysis of Results & Discussion	25%	Impactful Analysis: Provides a deep,	Correct Analysis: Correctly analyzes the	Superficial Analysis: Analysis contains errors	No Analysis: Results are simply presented (if at	/ 25

		quantitative discussion of the results. - Clearly compares the "before" (e.g., noisy signal) and "after" (e.g., filtered signal) states using appropriate metrics (e.g., SNR, visual clarity). - Discusses trade-offs and limitations.	"before" and "after" states. - Graphs are clear and discussed accurately. - Validation is present but may be superficial (e.g., "it sounds better" without metrics).	or is incomplete. - Graphs are missing, unlabeled, or misinterpreted. - Little to no comparison of "before" vs. "after."	all) without any interpretation or discussion of the application's success.	
4. Presentation & Live Demonstration	20%	Polished & Persuasive: - Presentation is clear, engaging, and well-timed. - Live demo is smooth and effectively	Clear & Functional: - Presentation covers all main points. - Live demo works.	Unclear: - Presentation is disorganized or difficult to follow. - Demo has technical issues.	Poor: - Presentation is incoherent. - Demo fails completely.	/ 20

		<i>sells</i> the solution. • Confidently answers all questions, demonstrating mastery of both the problem and the S&S theory.	• Answers most questions adequately.	• Struggles to answer questions.	• Unable to answer basic questions.	
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